NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

DEPARTMENT OF
COMPUTER SCIENCE

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PRIVACY-ENHANCED ONTOLOGIES

A PRACTICAL SSE M/M FRAMEWORK

MASTER IN COMPUTER SCIENCE

NOVA University Lisbon
Draft: *February 8, 2022*

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Abstract

1. What is the problem?
2. Why is this problem interesting/challenging?
3. What is the proposed approach/solution?
4. What results (implications/consequences) from the solution?

Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4

Resumo

1. Qual é o problema?
2. Porque é que é um problema interessante/desafiante?
3. Qual é a proposta de abordagem/solução?
4. Quais são as consequências/resultados da solução proposta?

Palavras-chave: Palavra-chave 1, Palavra-chave 2, Palavra-chave 3, ...

Contents

1

Introduction

1.1 Context

In 2001, the term *Semantic Web*[2] was first used to define what would be the evolution of the *World Wide Web*. The traditional *Web* solved the challenge of distributing information at scale and globally, however it was designed for human use. Documents are easily accessible and readable by humans, but for automated agents little more than document retrieval is possible. Contrary to humans, machines cannot reason over the information they access, if they do not know the *semantics of the data*.

The *Semantic Web* addresses this issue, by redesigning the representation of information, in a way that not only preserves meaning and relationships but is also machine-readable. A concept is mapped to a unique identifier, and meaning is encoded in sets of triples, each triple consisting of a subject, predicate and object (similar to an elementary sentence). *Ontologies* provide a shared understanding of a domain by formalizing the specification of relevant concepts, and their relations, in such domain.

This novel method of knowledge representation differs from the previous, centralized way, and greatly simplifies the process of interconnecting different knowledge bases. Centralized representations tend to be more expensive to develop, with almost every knowledge base needing to be built from scratch[7]. Furthermore, for sharing, both parties need to agree on the semantics. In contrast, using ontologies, information is semantically linked by default. In the *Semantic Web*, machines can truly communicate, share and reason about distinct knowledge bases.

1.2 Problem

Knowledge representation is essential in the industry[8] and ontologies help remove ambiguity in terminology. Nowadays, they are being developed and applied in a variety of domains[12] ranging from the health sector[1], to linguistics [2] and even being used by companies like NASA [1]. [3]

To developed and maintain ontologies, advanced semantic editors are the standard tool for professionals, with Protégé [6] being a very popular example. These provide numerous functionalities, usually based on plugins (e.g. automatic verification of errors and consistency in the design; debugging tools; query answering; graphic visualization of data).

With ontologies gradually growing in dimension, the industry is currently evolving towards the use of collaborative solutions [11, 10] and outsourcing heavy computations, relying on cloud infrastructures or other shared computing platforms. These approaches, albeit inevitable, do come with some new security concerns and engineering challenges.

1.3 Objectives

Briefly explain proposed solution. The scope of the solution, contributions, and how will it solve the problems mentioned in previous section.

1.4 Contributions

Enumerate the contributions.

1.5 Document Organization

Specify the document organization, per chapter.

[1] NCI Thesaurus[9], developed by the National Cancer Institute (NCI) or the Gene Ontology (GO)[3], the world's largest source of information on the functions of genes, funded by the National Human Genome Research Institute.

[2] WordNet[5], a lexical database for English where related words and concepts are semantically linked.

[3] Observing the visual representation provided by the Linked Open Data Cloud[4] is a good exercise to perceive the dimension of adoption of this new paradigm.

2

Related Work

2.1 Semantic Web

2.1.1 Languages

2.1.1.1 RDF: Resource Description Framework

2.1.1.2 RDF Schema

2.1.1.3 OWL2: Web Ontology Language

2.1.2 Querying the Semantic Web

2.1.2.1 RDF Stores

2.1.2.2 SPARQL

2.1.2.3 SWRL

2.1.3 Tools & Development Software for Ontologies

2.1.3.1 Semantic Editors

2.1.3.2 Semantic Reasoners

2.2 Searchable Encryption

2.2.1 Security Definitions

2.2.2 Client/Server Model

2.2.3 Symmetric Searchable Encryption

2.2.4 Homomorphic Encryption

2.3 Summary

3

Approach to Elaboration

3.1 System Requirements

3.2 Privacy Guaranties

3.3 Architecture & System Model

3.3.1 Implementation Guidelines

3.3.2 Validation

4

Work Plan

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